

PARIKSHIT MIT — TASK 1

Satellite Subsystems, Team Structure & Learning

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Subsystem: Administration & Operations.



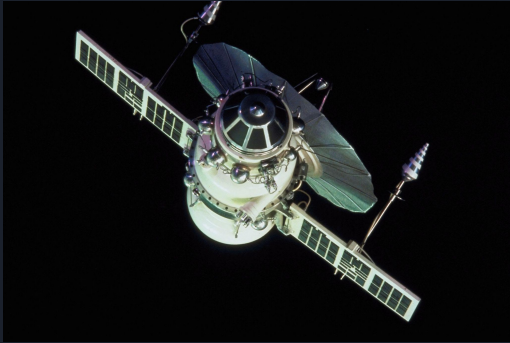
WHAT IS PARIKSHIT MIT?



PARIKSHIT is a student satellite project at MIT Manipal.

- Students work together to design and develop a satellite.
- The project gives students practical experience in satellite and space technology.
- Different teams work on different parts of the satellite.
- It helps students learn teamwork, engineering and project management.

PARIKSHIT SATELLITE

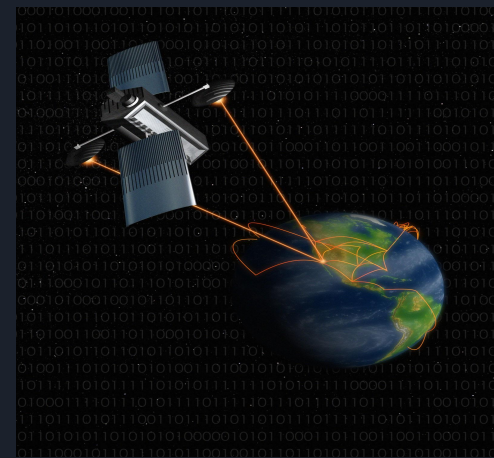
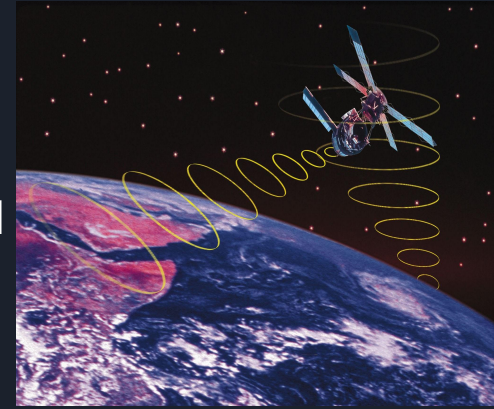


- Student satellite project at MIT Manipal
- Built and worked on by students
- Different teams handle different parts
- All the satellite systems have to work together
- Gives practical experience in satellite technology

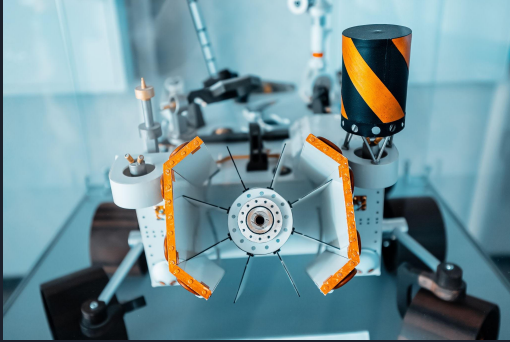


PARIKSHIT Teams & Subsystems

- ADCS – Controls the satellite's orientation and position
- COMMS – Handles communication between satellite and ground station
- EPS – Manages and supplies electrical power to the satellite
- ODHS – Handles onboard data and hardware systems
- PAYLOAD – Performs the main mission-related task
- STMS – Works on the satellite's structure and mechanical systems
- ADMIN & OPERATIONS – Handles coordination, planning and project activities.



PARIKSHIT – 2U CubeSat



- 2U CubeSat – small satellite based on the CubeSat standard
- 2U means two CubeSat units joined together
- Size is approximately $10 \times 10 \times 20$ cm
- Planned orbit – about 500 km above Earth
- Polar LEO – Low Earth Orbit with a near-polar path around Earth
- Designed to operate in Low Earth Orbit (LEO)
- The polar orbit allows the satellite to pass over different regions of Earth

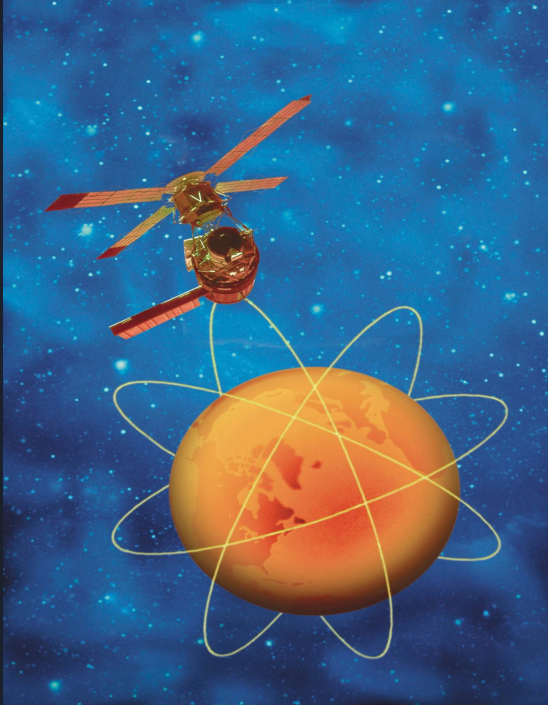


COMMS – Communication Subsystem

- Connects the satellite with the ground station
- Receives commands from Earth
- Sends satellite data back to Earth
- Uses antennas to transmit and receive signals
- Important for monitoring and controlling the satellite
- Helps send payload data to the ground station



PAYLOAD – Main Mission Equipment



- Payload is the main equipment used to perform the satellite's mission
- It collects the required information or performs the planned task
- Payload depends on the mission of the satellite
- It uses power and data from other satellite systems
- The collected payload data is sent to the ground through COMMS
- Payload is the main reason why the satellite is sent into space

ADMIN & OPERATIONS (AO)

- Coordinates and connects all the technical teams
- Plans and tracks project activities and deadlines
- Maintains important project documents and records
- Organises meetings, discussions and team communication
- Supports smooth execution of the satellite project
- Helps ensure work is completed in a planned and organised way



-A&O



- Recruitment and onboarding of members
- Fundraising and sponsorship activities
- Outreach, events & project promotion
- Sponsor and external organisation coordination
- Logistics, budgeting and the official correspondence
- Builds the project's public image and professional presence
- Helps maintain relationships with alumni, industry contacts and the wider community.



CONCLUSIONS

- Gained a basic understanding of the PARIKSHIT 2U CubeSat
- Learned about the major satellite subsystems and their functions
- Understood how different teams work together for one mission
- Learned the importance of planning, communication and teamwork
- Understood the role of Admin & Operations in supporting the project

